

avengeHERs

Meet Our Team



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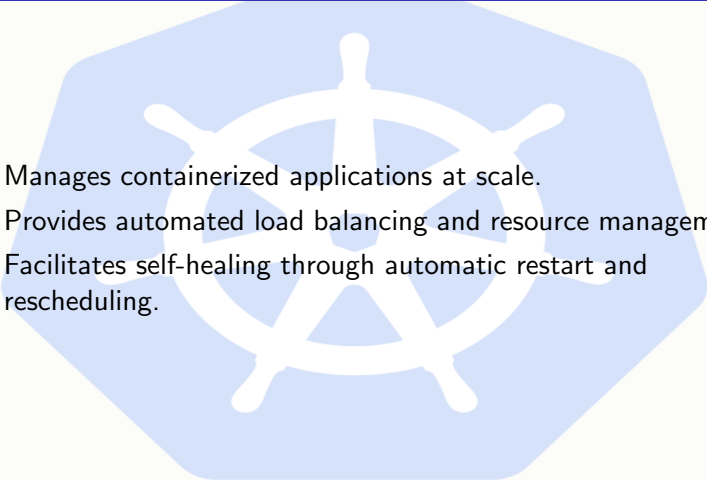
What is Kubernetes?

- Open-source platform for managing containerized applications.
- Developed by Google, now maintained by CNCF.
- Automates deployment, scaling, and management of containerized apps.

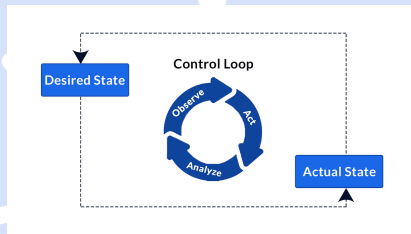
What are Containerized Applications?

- Applications packaged with all dependencies.
- Ensures consistency across different environments.
- Commonly used containers: Docker, CRI-O, containerd.

Why Kubernetes?

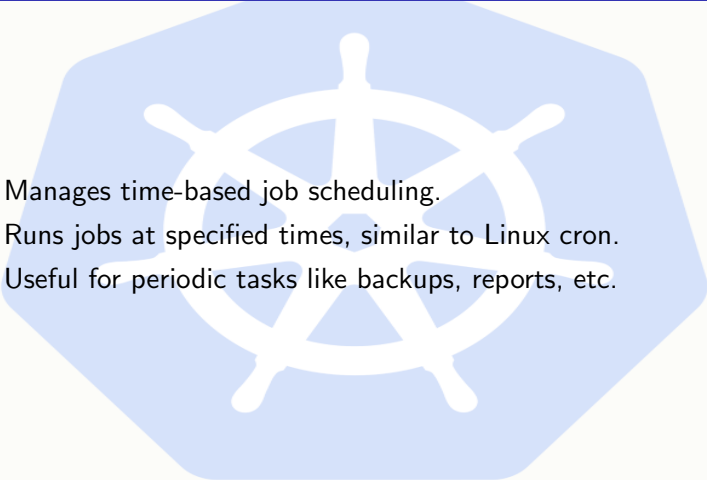
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- Manages containerized applications at scale.
 - Provides automated load balancing and resource management.
 - Facilitates self-healing through automatic restart and rescheduling.

Kubernetes Controllers

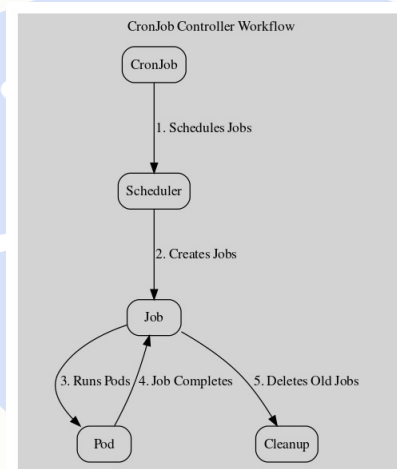


- Control loops that manage the state of the system.
- Ensure that the desired state matches the actual state.
- Examples: Deployment Controller, ReplicaSet Controller, CronJob Controller.

What is a CronJob Controller?

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- The Kubernetes logo, which is a white ship's wheel (helm) centered on a light blue octagonal background, is positioned behind the list items.
- Manages time-based job scheduling.
 - Runs jobs at specified times, similar to Linux cron.
 - Useful for periodic tasks like backups, reports, etc.

How the CronJob Controller Works



How the CronJob Controller Works

- Watches for CronJob resources in the cluster.
- Reads the schedule defined in the CronJob.
- Creates a Job resource at the scheduled time.
- The Job resource runs the specified task by creating Pods.
- Marks the Job as successful or failed based on task completion.
- Cleans up old Job resources to manage cluster resources efficiently.

Steps we followed

1 Set Up Environment:

- Installed Kubernetes, Kubebuilder, and Go.
- Ensured all necessary tools were correctly configured.

2 Created a New Project:

- Used Kubebuilder to initialize a new project.
- Scaffolded the basic structure of the controller.

3 Defined the CustomResourceDefinition (CRD):

- Created a CRD for CronJob.
- Specified the API schema for the CronJob resource.

4 Implemented Controller Logic:

- Developed the reconciliation logic to manage CronJobs.
- Handled events like creation, updates, and deletions.

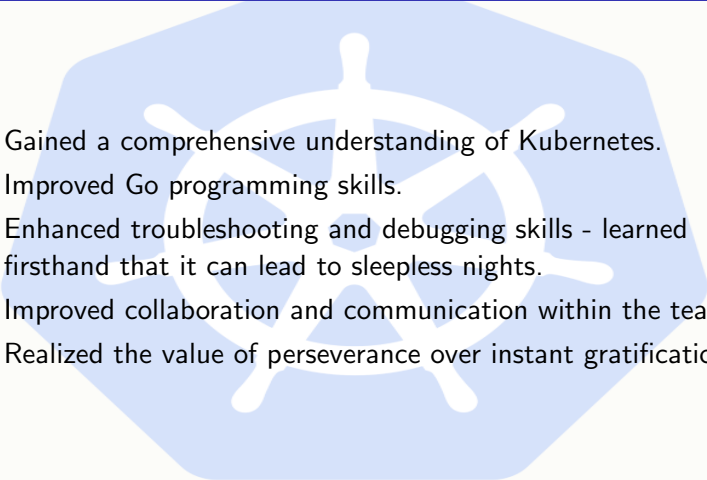
5 Tested and Validated:

- Ensured the controller ran and managed tasks correctly.

Challenges Faced

- Issues integrating the controller into the cluster.
- Go environment uninstalls itself when the laptop is powered off, causing interruptions.
- Confusing documentation made it challenging to follow the setup and usage.
- The make command resulted in errors.

Key Learnings

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- The background of the slide features a large, light blue octagon with a white ship's wheel (Kubernetes logo) in the center.
- Gained a comprehensive understanding of Kubernetes.
 - Improved Go programming skills.
 - Enhanced troubleshooting and debugging skills - learned firsthand that it can lead to sleepless nights.
 - Improved collaboration and communication within the team.
 - Realized the value of perseverance over instant gratification.

```
fmt.Println("Thank You")
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